B3.1 Gas exchange

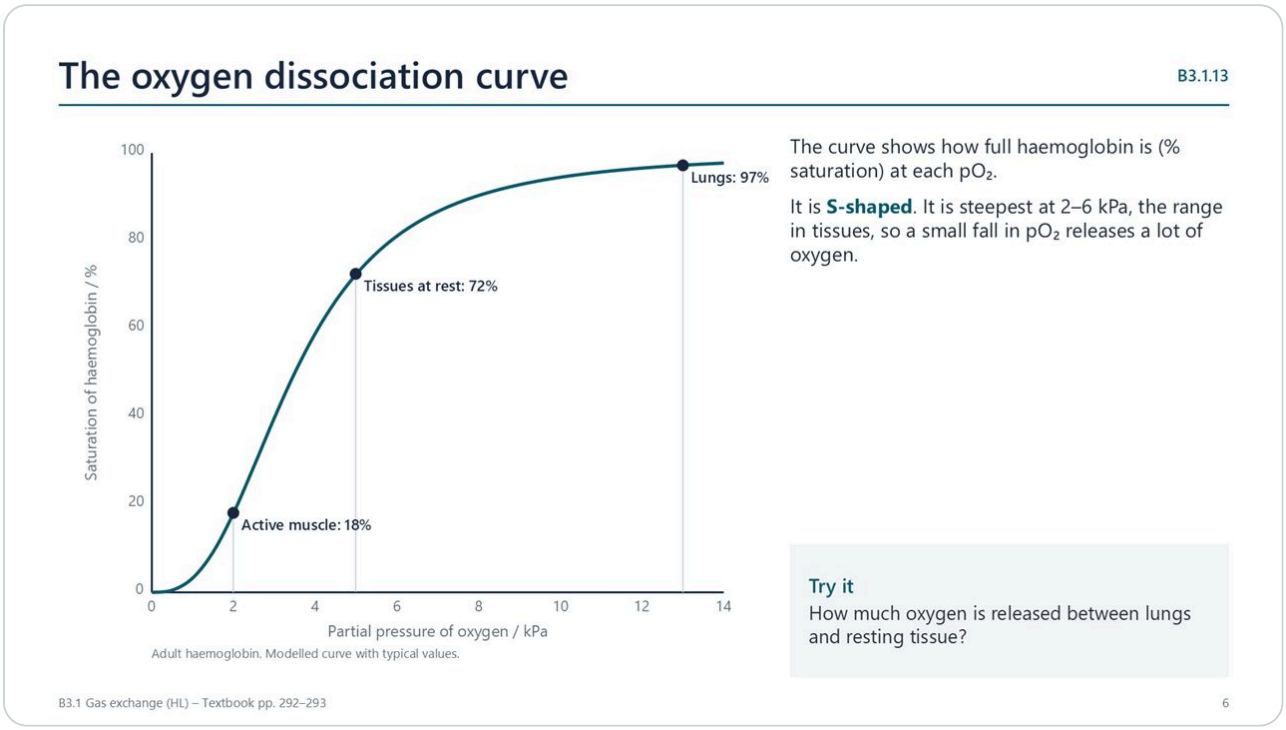
Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-b3-1.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 292–294 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

HAEMOGLOBIN AND OXYGEN

B3.1.13 Oxygen dissociation curves

pp. 292–293



The curve plots percentage saturation of haemoglobin against the **partial pressure of oxygen** (pO_2, in kPa). It is S-shaped because of **cooperative binding**: when one O_2 binds, the subunit changes shape and the other haem groups gain affinity. So haemoglobin loads almost fully in the lungs (about 13 kPa) and unloads readily in tissues (2–5 kPa), where the curve is steepest.

B3.1.11 Foetal and adult haemoglobin

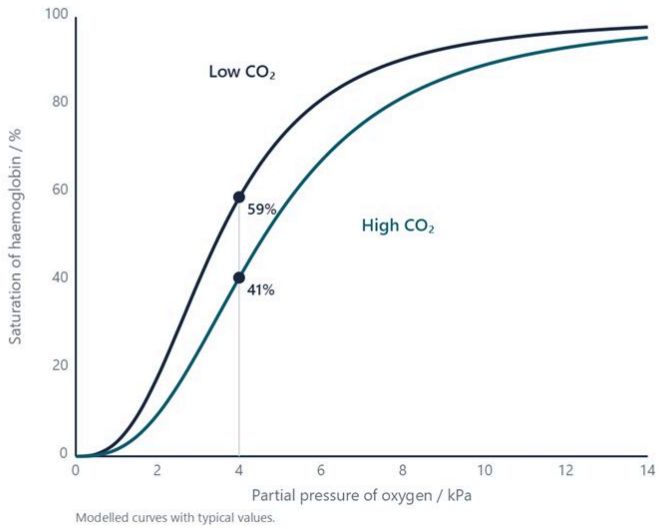
p. 293

Foetal haemoglobin has a **higher affinity** for oxygen, so its curve lies to the left. In the placenta, at the same pO_2, the mother's haemoglobin unloads while the foetus's loads. Haemoglobin is an allosteric protein: CO_2 binding at other sites lowers its affinity, and this effect is weaker in foetal haemoglobin. After birth it is replaced by the adult form.

B3.1.12 The Bohr shift

Reading the Bohr shift

B3.1.12



Modelled curves with typical values.

At 4 kPa: low CO_2 59%, high CO_2 41%. So an extra **18%** of the oxygen is released in tissue rich in CO_2.

At high pO_2 the curves almost meet, so loading in the lungs is hardly affected.

B3.1 Gas exchange (HL) – Textbook p. 294

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In respiring tissue, CO_2 forms carbonic acid in red blood cells, releasing H^+ and lowering pH. H^+ and CO_2 change the shape of haemoglobin, so its affinity falls and the curve shifts **right**. More oxygen is released exactly where respiration is fastest; in the lungs CO_2 leaves and affinity rises again.

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